

A List of Research Topics

We selected the following list of research topics for our research ideation task:

1. Bias: novel prompting methods to reduce social biases and stereotypes of large language models
2. Coding: novel prompting methods for large language models to improve code generation
3. Safety: novel prompting methods to improve large language models' robustness against adversarial attacks or improve their security or privacy
4. Multilingual: novel prompting methods to improve large language models' performance on multilingual tasks or low-resource languages and vernacular languages
5. Factuality: novel prompting methods that can improve factuality and reduce hallucination of large language models
6. Math: novel prompting methods for large language models to improve mathematical problem solving
7. Uncertainty: novel prompting methods that can better quantify uncertainty or calibrate the confidence of large language models

We use these topics descriptions to elicit ideas from both human participants and our LLM agent.

B Project Proposal Template

We give the following project proposal template to both the AI agent and human idea writers.

- 1. Title:** A concise statement of the main research question to be used as the paper title.
- 2. Problem Statement:** Clearly define the problem your research intends to address. Explain clearly why this problem is interesting and important.
- 3. Motivation:** Explain why existing methods are not good enough to solve the problem, and explain the inspiration behind the new proposed method. You should also motivate why the proposed method would work better than existing baselines on the problem.
- 4. Proposed Method:** Explain how the proposed method works, describe all the essential steps.
- 5. Step-by-Step Experiment Plan:** Break down every single step of the experiments, make sure every step is executable. Cover all essential details such as the datasets, models, and metrics to be used. If the project involves prompting, give some example prompts for each step.
- 6. Test Case Examples:** Give at least two concrete examples. The first example should show how the baseline method fails on the test case. If there are multiple baselines, give examples for all of them. The second example should show how the proposed method succeeds on the test case. For each test case, include the input (test example and the full prompt) and the expected output. You should also provide an explanation for why the outputs from the proposed prompt are better. If the proposed method has multiple steps, break them down into intermediate steps.
- 7. Fallback Plan:** Propose some alternative plans for what should the students do if the proposed method doesn't manage to satisfy the success criteria. For example, you can suggest additional analysis to help debug why the proposed method didn't work, which could inform alternative new methods, or just turn the project into an analysis paper instead by offering some interesting ablation and insights.

C Project Proposal Demo Example

We present a manually written demonstration example used for project proposal generation. The example is summarized from an existing paper (Dhuliawala et al., 2023). This same example is given to both the AI agent as well as the idea-writing experts.

1. Title:

Chain-of-Verification Reduces Hallucination in Large Language Models

2. Problem Statement:

Generation of plausible yet incorrect factual information, termed hallucination, is an unsolved issue in large language models.

3. Motivation:

A majority of the methods for reducing hallucination can be divided into roughly three categories: training-time correction, generation-time correction, and via augmentation (tool-use). We want to take a simpler approach that fully leverages the power of LLM itself. Our key motivation is that large language models, when suitably prompted, can both generate and execute a plan of how to verify themselves in order to check their own work, and finally incorporate this analysis into an improved response.

4. Proposed Method:

Our overall process, which we call Chain-of-Verification (CoVe), thus performs four core steps:

- (1) **Generate Baseline Response:** Given a query, generate the response using the LLM.
- (2) **Plan Verifications:** Given both query and baseline response, generate a list of verification questions that could help to self-analyze if there are any mistakes in the original response.
- (3) **Execute Verifications:** Answer each verification question in turn, and hence check the answer against the original response to check for inconsistencies or mistakes.
- (4) **Generate Final Verified Response:** Given the discovered inconsistencies (if any), generate a revised response incorporating the verification results.

Each of these steps is performed by prompting the same LLM in different ways to obtain the desired response.

5. Step-by-Step Experiment Plan:

- 1: **Gather Datasets:** We choose datasets that evaluate factual correctness, including the Multi-SpanQA dataset on closed-book QA and the FactScore dataset on generating biographies.
- 2: **Construct Prompts:** For the baseline, we use direct prompting where, given a query, we generate left-to-right as usual using the LLM, with no special tricks. Given that such baseline generations are typically prone to hallucination, CoVe attempts to identify these hallucinations and correct them in the following steps:
 - (1) **Plan Verifications:** Conditioned on the original query and the baseline response, the model is prompted to generate a series of verification questions that test the factual claims in the original baseline response.
 - (2) **Execute Verifications:** Given the planned verification questions, the next step is to answer them in order to assess if any hallucinations exist. The planning prompt conditions on the baseline response in the first step. The verification questions generated from planning are answered in the second step, where crucially the context given to the LLM prompt only contains the questions and not the original baseline response, hence preventing the LLM from repeating those answers directly.

- (3) **Generate Final Verified Response:** Finally, the improved response that takes verification into account is generated. This is executed by a final few-shot prompt where the context takes into account all of the previous reasoning steps, the baseline response, and verification question-answer pairs, so that the corrections can take place.
- 3: **Select Models:** We test GPT-3.5 (Text-Davinci-003) and GPT-4 from the OpenAI API, as well as the open-source LLaMA-2-70B-chat.
- 4: **Get Results:** Get answer predictions from the models on these datasets with both the baselines and the proposed method.
- 5: **Analyze Results:** Compare whether the new method improves the performance of LLMs in these tasks as compared to the baselines.

6. Test Case Examples:

- **Baseline Prompt Input (Direct Prompting):** Name some politicians who were born in NY, New York.
- **Baseline Prompt Expected Output (Direct Prompting):** Here are some politicians who were born in NY, New York:
 1. Hillary Clinton - former secretary of state and former Democratic presidential nominee.
 2. Donald Trump - former president of the United States.
 3. Michael Bloomberg - former Mayor of New York City and former Democratic presidential candidate.
- **Proposed Prompt Input (Chain-of-Verification Prompting; Step 1: Plan Verifications):** Generate a set of verification questions to verify each answer.
- **Proposed Prompt Expected Output (Chain-of-Verification Prompting; Step 1: Plan Verifications):**
 - Where was Hillary Clinton born?
 - Where was Donald Trump born?
 - Where was Michael Bloomberg born?
- **Proposed Prompt Input (Chain-of-Verification Prompting; Step 2: Execute Verifications):** Answer each of the verification questions.
- **Proposed Prompt Expected Output (Chain-of-Verification Prompting; Step 2: Execute Verifications):**
 1. Hillary Clinton was born in Chicago, Illinois, United States on October 26, 1947.
 2. Donald Trump was born on June 14, 1946, in Queens, New York City, New York, United States.
 3. Michael Bloomberg was born on February 14, 1942, in Boston, Massachusetts, United States.
- **Proposed Prompt Input (Chain-of-Verification Prompting; Step 3: Final Response):** (Prepend the original question and the baseline response and the verification QA pairs) Given the verification answers, generate a revised response incorporating the verification results.
- **Proposed Prompt Expected Output (Chain-of-Verification Prompting; Step 3: Final Response):** Here are some politicians who were born in NY, New York:
 1. Donald Trump - former president of the United States.
 2. Alexandria Ocasio-Cortez - Democratic member of the U.S. House of Representatives.
- **Explanation:** Given a user query, a large language model with direct prompting generates a baseline response that may contain inaccuracies, e.g., factual hallucinations. To improve this, Chain-of-Verification first generates a plan of a set of verification questions to ask, and then

executes that plan by answering them and hence checking for agreement. We find that individual verification questions are typically answered with higher accuracy than the original accuracy of the facts in the original longform generation. Finally, the revised response takes into account the verifications.

7. Fallback Plan:

If the proposed method does not help as compared to the baseline, analyze each step of the CoVe process to see if the verification questions are relevant, if the answers to the verification questions are correct, and whether the generated final verified response is indeed improved over the baseline response by considering the verification QA pairs. This can help us debug the proposed method or turn this into interesting analysis on the model’s ability to verify and correct its own responses.